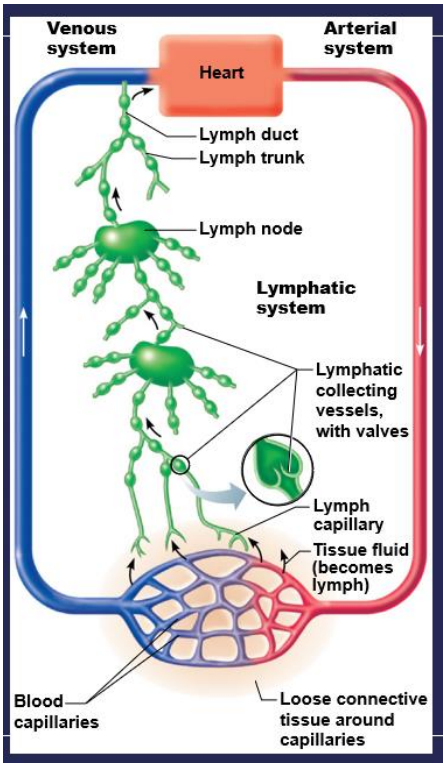
The Lymphatic System and Immunity

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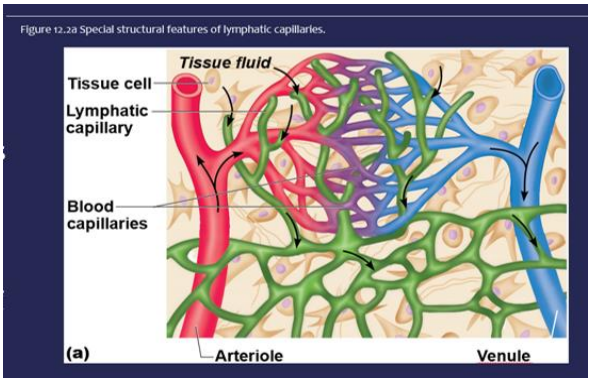
FUNCTIONS OF LYMPHATIC SYSTEM

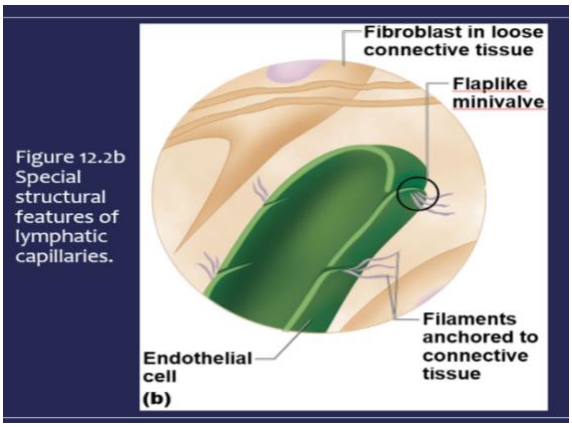
1. Fluid Balance
   - ✓ 30L of fluid pass from capillaries to interstitial spaces each day
   - ✓ 27 L pass from interstitial spaces back to blood capillaries
   - ✓ 3L enters lymphatic capillaries
   - ✓ Transports escaped fluids from the cardiovascular system back to the blood
2. Fat Absorption
   - ✓ absorbs fat in digestive tract- lacteals
3. Defense
   - ✓ Plays essential roles in body defense and resistance to disease



FUNCTIONS OF LYMPHATIC VESSELS

- Lymph
  - consists of excess tissue fluid and plasma proteins carried by lymphatic vessels
  - if fluids are not picked up, edema occurs as fluid accumulates in tissues
- Lymphatic vessels
  - pick up lymph and return it to the blood
  - form a one-way system
  - lymph flows only towards the heart
  - are similar to veins of the cardiovascular system
    - Thin-walled
    - Larger vessels have valves
    - Low-pressure, pumpless system
- Lymph transport is aided by:
  - Milking action of skeletal muscles
  - Pressure changes in thorax during breathing
  - Smooth muscle in walls of lymphatics
- Lymph capillaries
  - Weave between tissue cells and blood capillaries
  - Walls overlap to form flaplike mini valves
  - Fluid leaks into lymph capillaries
  - Capillaries are anchored to connective tissue by filaments
  - Higher pressure on the inside closes mini valves
  - Fluid is forced along the vessel

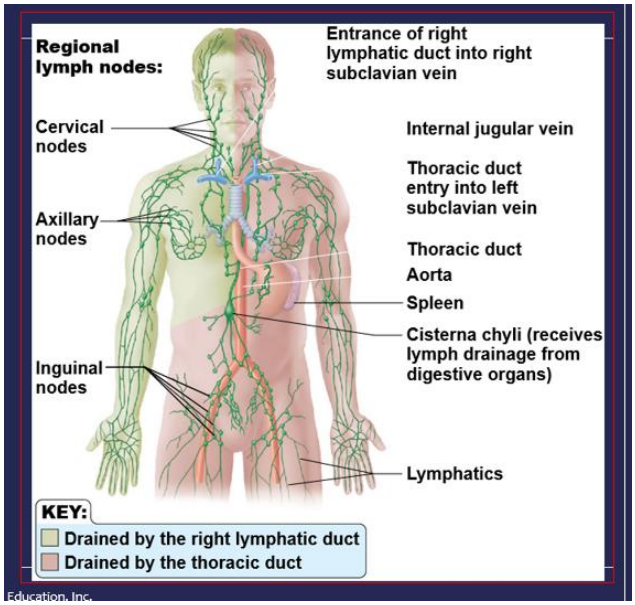




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❖ Lymphatic collecting vessels

- Collect lymph from lymph capillaries
- Carry lymph to and away from lymph nodes
- Return fluid to circulatory veins near the heart
- Right lymphatic duct drains the lymph from the right arm and the right side of the head and thorax
- Thoracic duct drains lymph from rest of body



LYMPHATIC ORGANS AND THEIR FUNCTIONS

1. Lymph Nodes
2. Other Lymphoid Organs
   a) Tonsils
   b) Thymus
   c) Spleen
   d) Peyer's patches
   e) Appendix

❖ LYMPH NODES

✓ Lymph nodes filter lymph before it is returned to the blood

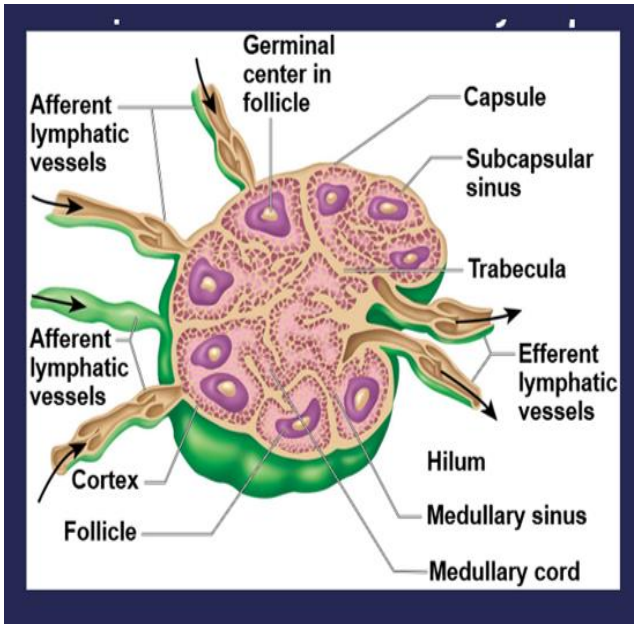
✓ Harmful materials that are filtered
- Bacteria
- Viruses
- Cancer cells
- Cell debris

✓ Defense cells within lymph nodes
- Macrophages—engulf and destroy bacteria, viruses, and other foreign substances in lymph
- Lymphocytes—respond to foreign substances in lymph

✓ Most lymph nodes are kidney-shaped, less than 1 inch long, and buried in connective tissue

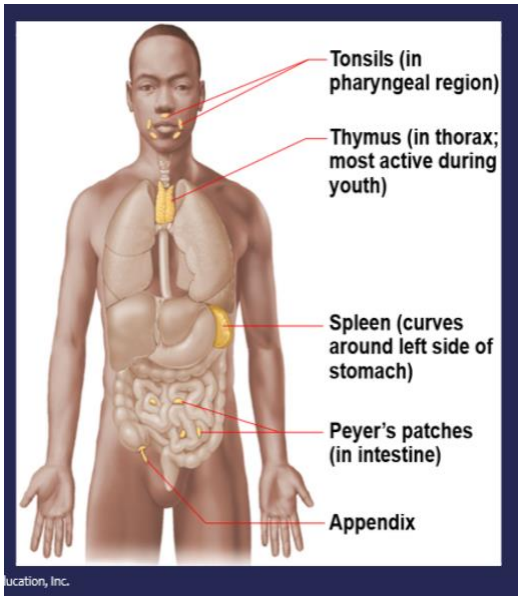
✓ Surrounded by a capsule

✓ Divided into compartments by trabeculae
- Cortex (outer part)
  - Contains follicles—collections of lymphocytes
  - Germinal centers enlarge when antibodies are released by plasma cells
- Medulla (inner part)
  - Contains phagocytic macrophages



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✓ Flow of Lymph through Nodes
- Lymph enters the convex side through afferent lymphatic vessels
- Lymph flows through a number of sinuses inside the node
- Lymph exits through efferent lymphatic vessels
- Because there are fewer efferent than afferent vessels, flow is slowed

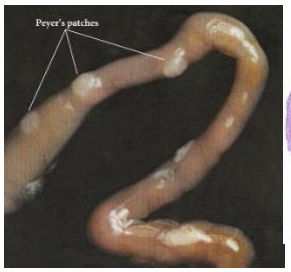


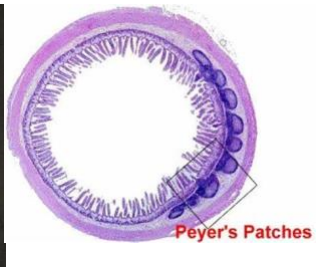
OTHER LYMPHATIC ORGANS

❖ TONSILS
✓ Small masses of lymphoid tissue deep to the mucosa surrounding the pharynx
✓ Trap and remove bacteria and other foreign pathogens
✓ Tonsillitis results when the tonsils become congested with bacteria

❖ THYMUS
✓ Found overlying the heart
✓ Produces thymosin (creates T-lymphocytes)
✓ Site of production and maturation of T lymphocytes

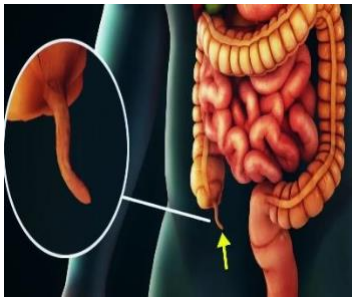
❖ SPLEEN
✓ Located on the left side of the abdomen
✓ Filters and cleans blood of bacteria, viruses, debris
✓ Provides a site for lymphocyte proliferation and immune surveillance
✓ Destroys worn-out blood cells
✓ Forms blood cells in the fetus
✓ Acts as a blood reservoir





❖ PEYER'S PATCH
✓ Found in the wall of the small intestine
✓ Similar lymphoid follicles are found in the appendix
✓ Macrophages capture and destroy bacteria in the intestine

❖ APPENDIX
✓ vestigial hollow tube that is closed at one end and is attached at the other end to the cecum
✓ housing and cultivating beneficial gut flora



Mucosa-associated lymphoid tissue (MALT)
Includes:
- Peyer's patches
- Tonsils
- Appendix

Acts as a sentinel to protect respiratory and digestive tracts

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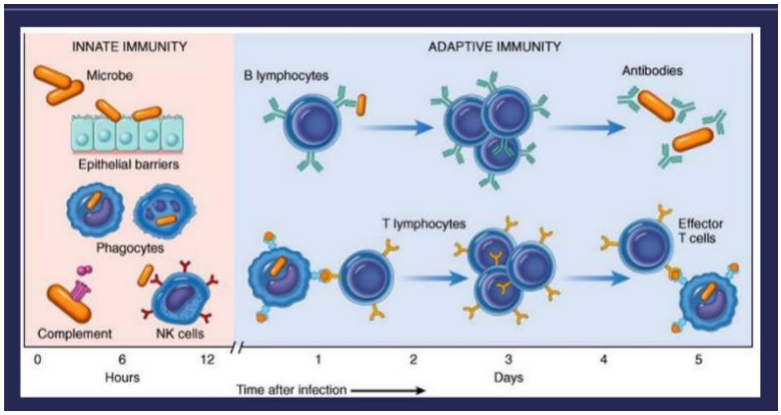
IMMUNITY AND ITS COMPONENTS

- Immunity
  - ✓ specific resistance to disease
- Immune system
  - ✓ is a *functional* system rather than an organ system in an anatomical sense
- Two mechanisms that make up the immune system defend us from foreign materials
  - ✓ Innate (non-specific) defense system
  - ✓ Adaptive (specific) defense system

The Immune System

Innate (non-specific) defense mechanisms		Adaptive (specific) defense mechanisms
First line of defense	Second line of defense	Third line of defense
• Skin • Mucous membranes • Secretions of skin and mucous membranes	• Phagocytic cells • Natural killer cells • Antimicrobial proteins • The inflammatory response • Fever	• Lymphocytes • Antibodies • Macrophages and other antigen-presenting cells

Innate (nonspecific) defense system	Adaptive (specific) defense system
• Mechanisms protect against a variety of invaders • Responds immediately to protect body from foreign materials • mechanical barriers to pathogens	• Fights invaders that get past the innate system • Specific defense is required for each type of invader • The highly specific resistance to disease is *immunity*



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Table **12.1** **Summary of Innate (Nonspecific) Body Defenses**

Category and associated elements	Protective mechanism
Surface membrane barriers—first line of defense	
Intact skin (epidermis)	Forms mechanical barrier that prevents entry of pathogens and other harmful substances into body.
• Acid mantle	Skin secretions make epidermal surface acidic, which inhibits bacterial growth; sebum also contains bacteria-killing chemicals.
• Keratin	Provides resistance against acids, alkalis, and bacterial enzymes.
Intact mucous membranes	Form mechanical barrier that prevents entry of pathogens.
• Mucus	Traps microorganisms in respiratory and digestive tracts.
• Nasal hairs	Filter and trap microorganisms and other airborne particles in nasal passages.
• Cilia	Propel debris-laden mucus away from lower respiratory passages.
• Gastric juice	Contains concentrated hydrochloric acid and protein-digesting enzymes that destroy pathogens in stomach.
• Acid mantle of vagina	Inhibits growth of bacteria and fungi in female reproductive tract.
• Lacrimal secretion (tears); saliva	Continuously lubricate and cleanse eyes (tears) and oral cavity (saliva); contain lysozyme, an enzyme that destroys microorganisms.

Table **12.1** **Summary of Innate (Nonspecific) Body Defenses** (*continued*)

Category and associated elements	Protective mechanism
Cellular and chemical defenses—second line of defense	
Phagocytes	Engulf and destroy pathogens that breach surface membrane barriers; macrophages also contribute to immune response.
Natural killer cells	Promote cell lysis by direct cell attack against virus-infected or cancerous body cells; do not depend on specific antigen recognition.
Inflammatory response	Prevents spread of injurious agents to adjacent tissues, disposes of pathogens and dead tissue cells, and promotes tissue repair; releases chemical mediators that attract phagocytes (and immune cells) to the area.

Table **12.1** **Summary of Innate (Nonspecific) Body Defenses** (*continued*)

Category and associated elements	Protective mechanism
Cellular and chemical defenses—second line of defense	
Antimicrobial chemicals	
• Complement	Group of plasma proteins that lyses microorganisms, enhances phagocytosis by opsonization, and intensifies inflammatory response.
• Interferons	Proteins released by virus-infected cells that protect uninfected tissue cells from viral takeover; mobilize immune system.
• Fluids with acid pH	Normally acid pH inhibits bacterial growth; urine cleanses the lower urinary tract as it flushes from the body.
Fever	Systemic response triggered by pyrogens; high body temperature inhibits multiplication of bacteria and enhances body repair processes.

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- Inflammatory Response
  - ✓ Triggered when body tissues are injured
  - ✓ Four most common indicators (cardinal signs) of acute inflammation
    - Redness
    - Pain
    - Swelling (edema)
    - Heat
  - ✓ Damaged cells release inflammatory chemicals
    - Histamine
    - Kinin
  - ✓ These chemicals cause:
    - Blood vessels to dilate
    - Capillaries to become leaky
    - Phagocytes and white blood cells to move into the area (called *positive chemotaxis*)

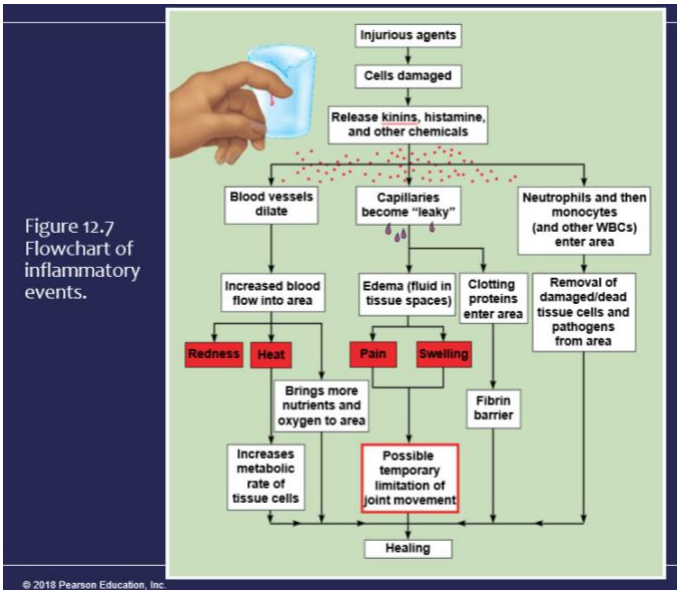
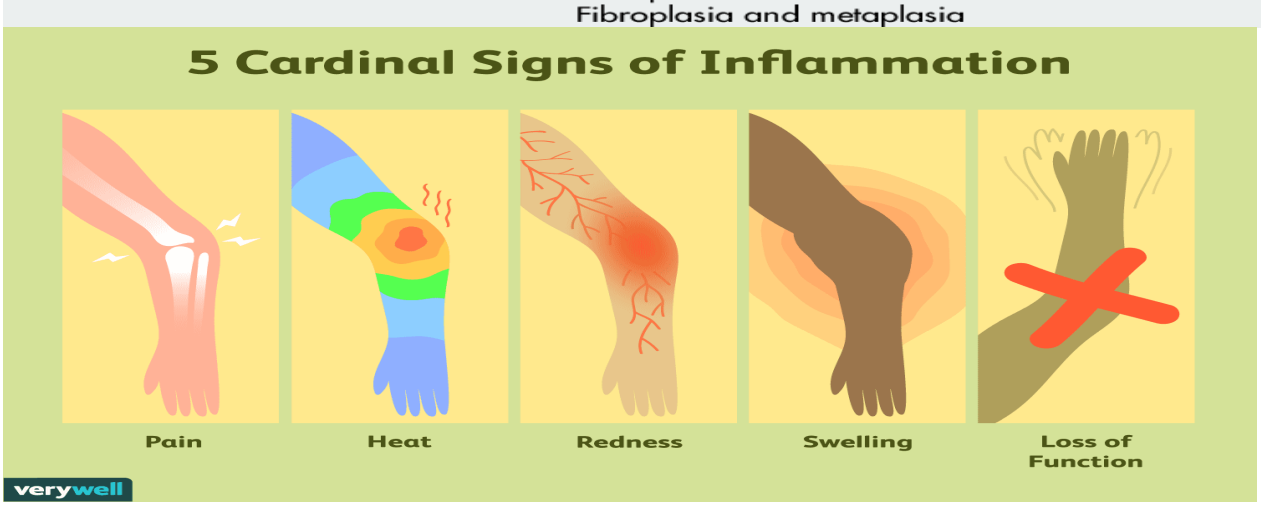


Figure 12.7 Flowchart of inflammatory events.

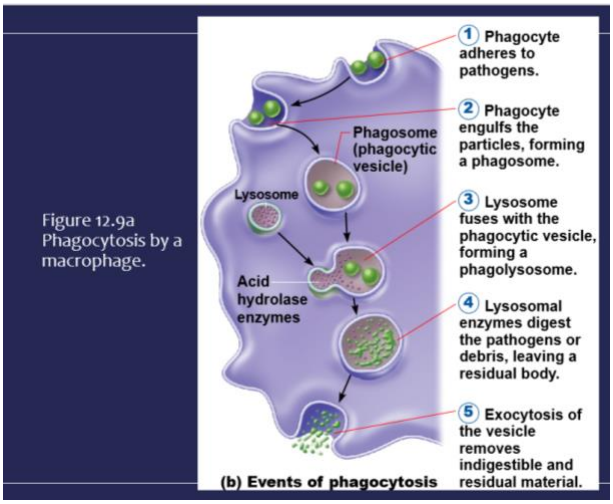
English	Latin	Cause
Heat	*Calor*	Vasodilation
Redness	*Rubor*	Vasodilation
Swelling	*Tumor*	Increased vascular permeability
		Increased granulation tissue
Pain	*Dolor*	Physical and chemical stimulation of nociceptors
Loss of function	*Functio laesa*	Pain
		Reflex muscle inhibition
		Disruption of tissue structure
		Fibroplasia and metaplasia

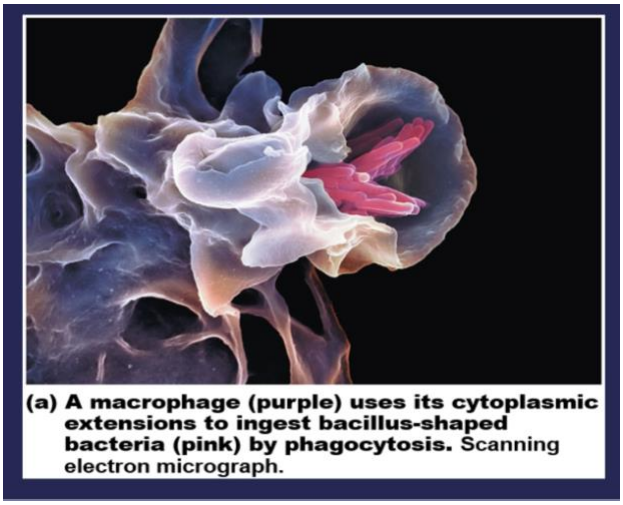


- Functions of the Inflammatory Response
  - Prevents spread of damaging agents
  - Disposes of cell debris and pathogens through phagocytosis
  - Sets the stage for repair
- Process of the Inflammatory Response
  - Neutrophils migrate to the area of inflammation by rolling along the vessel wall (following the scent of chemicals from inflammation)
  - Neutrophils squeeze through the capillary walls by diapedesis to sites of inflammation

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- Neutrophils gather in the precise site of tissue injury (positive chemotaxis) and consume any foreign material present
- Phagocytes
- Cells such as neutrophils and macrophages engulf foreign material by phagocytosis
- The phagocytic vesicle is fused with a lysosome, and enzymes digest the cell's contents





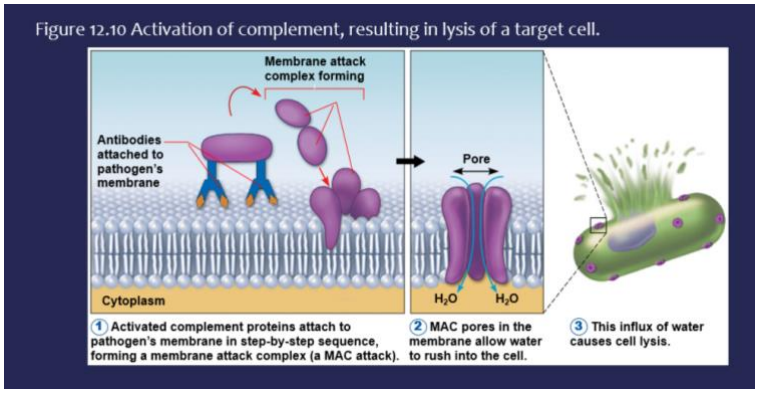
❖ Antimicrobial proteins
- Enhance innate defenses by:
  - ✓ Attacking microorganisms directly
  - ✓ Hindering reproduction of microorganisms
- Most important types
  - ✓ Complement proteins
  - ✓ Interferon

➢ Complement Proteins

✓ *Complement* refers to a group of at least 20 plasma proteins that circulate in the plasma

✓ Complement is activated when these plasma proteins encounter and attach to cells known as *complement fixation*

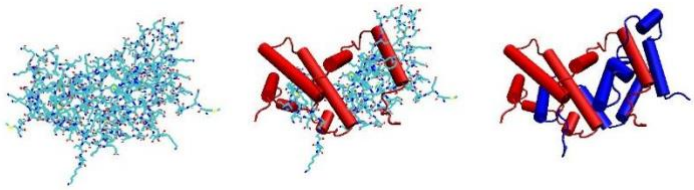
✓ MEMBRANE ATTACK COMPLEXES (MACs), one result of complement fixation, produce holes or pores in cells
- Pores allow water to rush into the cell
- Cell bursts (lyses)
- Activated complement enhances the inflammatory response



➢ Interferons

✓ Interferons are small proteins secreted by virus-infected cells

✓ Interferons bind to membrane receptors on healthy cell surfaces to *interfere* with the ability of viruses to multiply



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❖ Fever

✓ Abnormally high body temperature is a systemic response to invasion by microorganisms

✓ Hypothalamus regulates body temperature at 37ºC (98.6ºF)

✓ The hypothalamus thermostat can be reset higher by pyrogens (secreted by white blood cells)

✓ High temperatures inhibit the release of iron and zinc from the liver and spleen (needed by bacteria)

✓ Fever also increases the speed of repair processes

ADAPTIVE BODY DEFENSES

✓ Adaptive body defenses are the body's specific defense system, or the *third line of defense*

✓ Immune response is the immune system's response to a threat

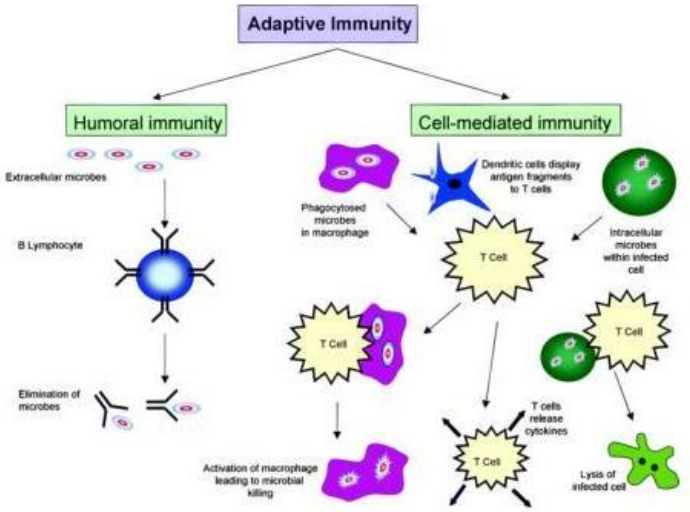
✓ Antigens are targeted and destroyed by antibodies

✓ Three aspects of adaptive defense

- Antigen specific—the adaptive defense system recognizes and acts against particular foreign substances
- Systemic—immunity is not restricted to the initial infection site
- Memory—the adaptive defense system recognizes and mounts a stronger attack on previously encountered pathogens

✓ Two arms of the adaptive defense system

- Humoral immunity = antibody-mediated immunity
  -Provided by antibodies present in body fluids
- Cellular immunity = cell-mediated immunity
  - Targets virus-infected cells, cancer cells, and cells of foreign grafts



ANTIGENS

Antigens are any substance capable of exciting the immune system and provoking an immune response

COMMON NON-SELF ANTIGENS

✓ Foreign proteins provoke the strongest response

- Pollen grains
- Microorganisms (bacteria, fungi, viruses)

SELF-ANTIGENS

✓ Self-antigens do not trigger an immune response in us

✓ The presence of our cells in another person's body can trigger an immune response because they are foreign

✓ Restricts donors for transplants

HAPTENS, OR *INCOMPLETE ANTIGENS*

✓ are not antigenic by themselves

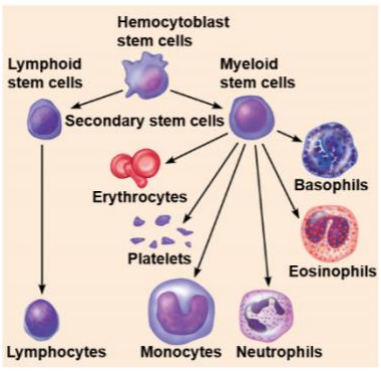
✓ when they link up with our own proteins, the immune system may recognize the combination as foreign and respond with an attack

- found in poison ivy, animal dander, detergents, hair dyes, cosmetics

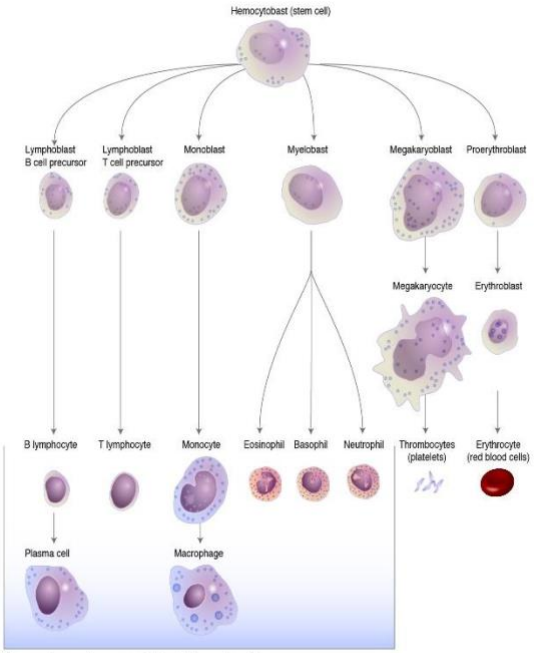
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CRUCIAL CELLS OF THE ADAPTIVE SYSTEM

- ❖ Lymphocytes
  - ✓ respond to specific antigens
  - ✓ arise from hemocytoblasts of bone marrow
    - B lymphocytes (B cells)
    - T lymphocytes (T cells)
- ❖ Immunocompetence
  - ✓ The capability to respond to a specific antigen by binding to it with antigen-specific receptors that appear on the lymphocyte's surface
  - ✓ Antigen-presenting cells (APCs)—help the lymphocytes but do not respond to specific antigens



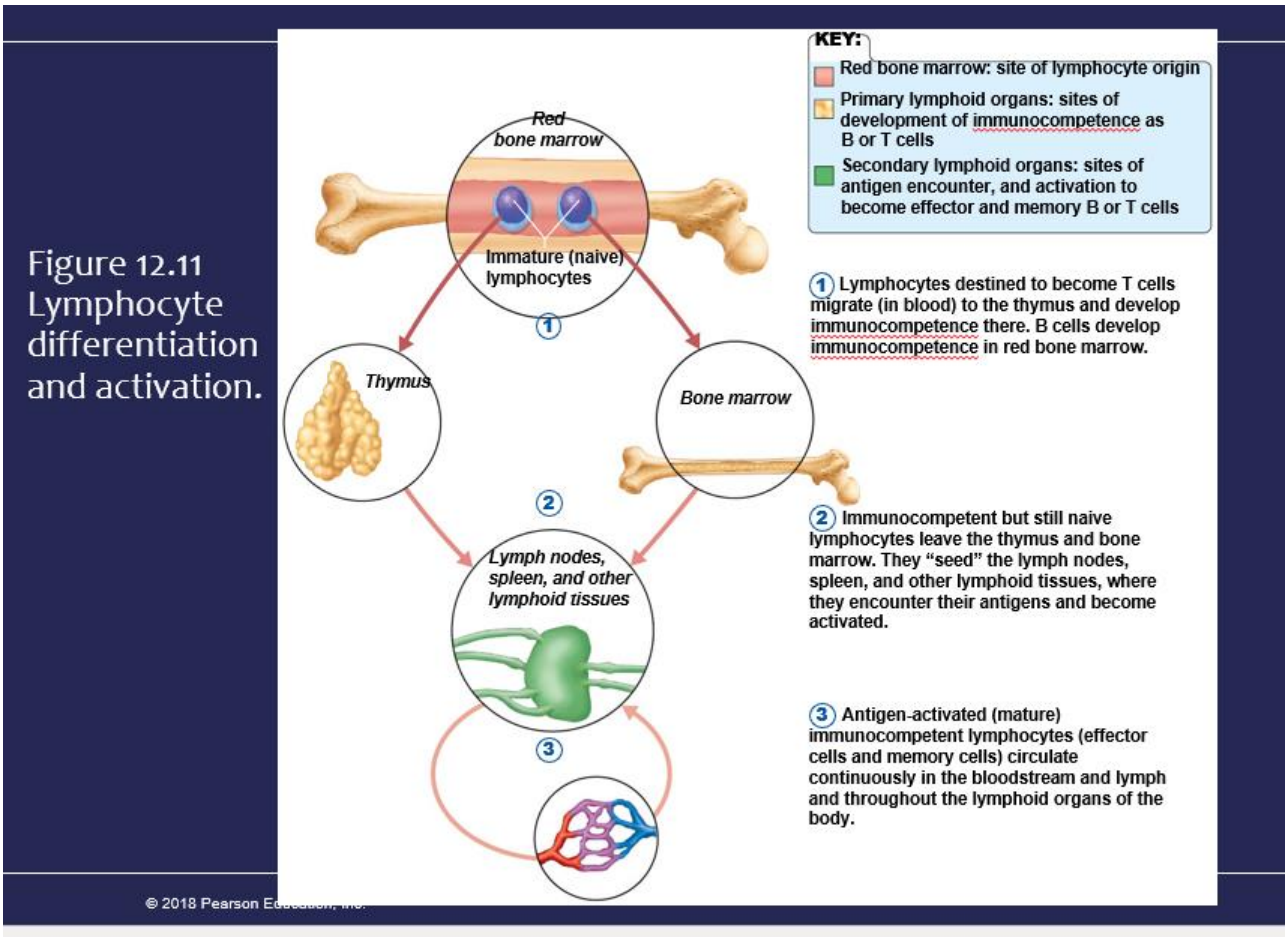
T cells Lymphocytes	B cells Lymphocytes
• develop immunocompetence in the thymus and oversee cell-mediated immunity • Identify foreign antigens • Self-tolerance is important part of lymphocyte "education" • do not make antibodies • fight antigens directly	• develop immunocompetence in bone marrow • provide humoral immunity • Produce antibodies • Behave as APCs
Immunocompetent T and B lymphocytes migrate to the lymph nodes and spleen, where encounters with antigens occur	



Lymphocytes - white blood cells

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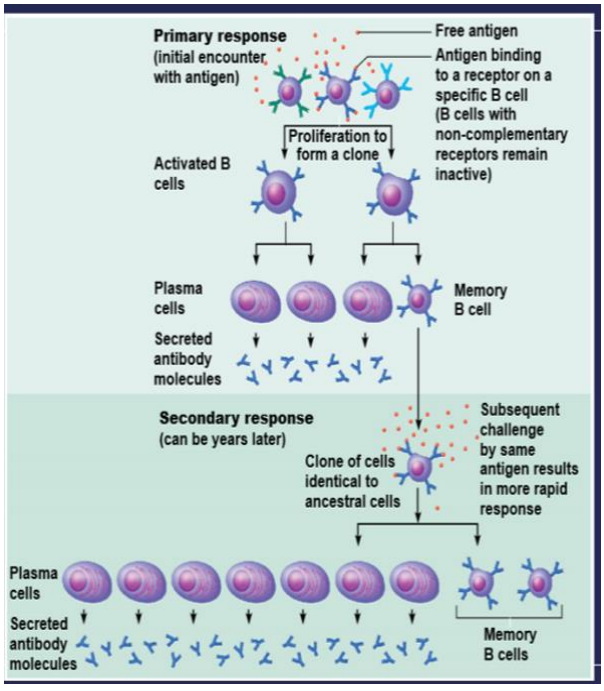
V



- ❖ Antigen-Presenting Cells (APCS)
- ✓ Engulf antigens and then present fragments of them on their own surfaces, where they can be recognized by T cells
- ✓ Major types of cells behaving as APCs
  - Dendritic cells
  - Macrophages
  - B lymphocytes
- ✓ When they present antigens, dendritic cells and macrophages activate T cells, which release chemicals

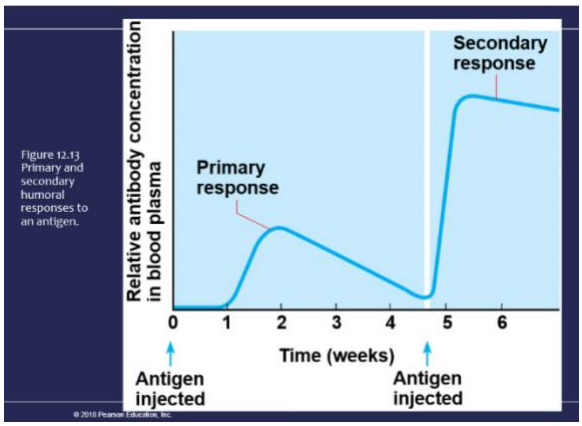
PRIMARY HUMORAL RESPONSE

- ✓ B lymphocytes with specific receptors bind to a specific antigen
- ✓ The binding event sensitizes, or activates, the lymphocyte to undergo clonal selection
- ✓ A large number of clones is produced
- ✓ Most of the B cell clone members (descendants) become plasma cells
- ✓ Produce antibodies to destroy antigens
- ✓ Activity lasts for 4 or 5 days
- ✓ Plasma cells begin to die



SECONDARY HUMORAL RESPONSE

- ✓ Some B cells become long-lived memory cells capable of mounting a rapid attack against the same antigen in subsequent meetings
- ✓ These cells provide immunological memory



❖ Humoral (Antibody-Mediated) Immune Response

✓ Antibodies (immunoglobulins, Igs)

✓ Constitute gamma globulin part of blood proteins

✓ Soluble proteins secreted by activated B cells (plasma cells)

✓ Formed in response to a huge number of antigens

❖ Antibody structure

✓ Four polypeptide chains, two heavy and two light, linked by disulfide bonds to form a T- or Y-shaped molecule

✓ Each polypeptide chain has a variable (V) region and a constant (C) region

✓ Variable regions form antigen-binding sites, one on each arm of the T or Y

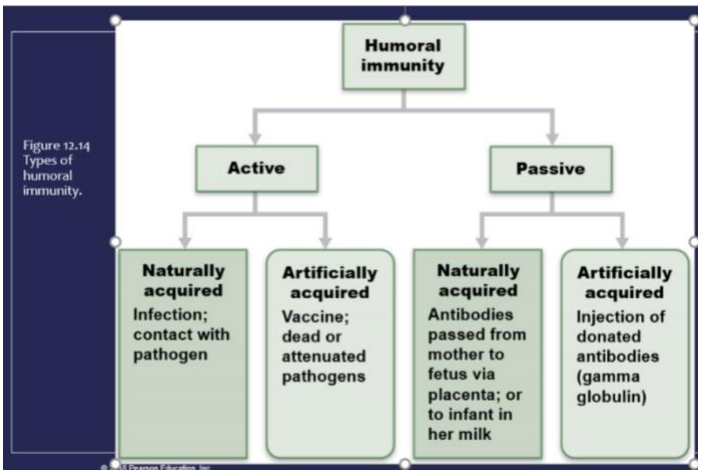
✓ Constant regions determine the type of antibody formed (antibody class)

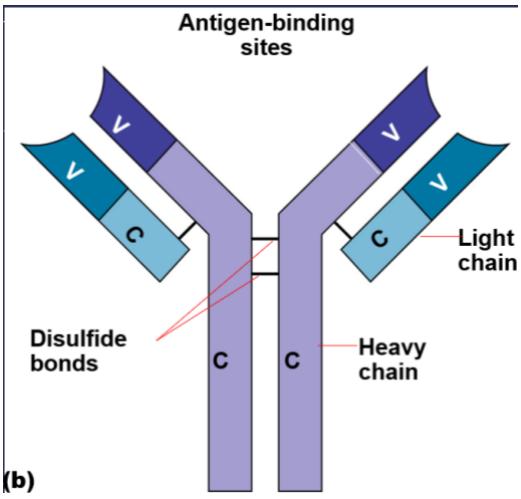
❖ Antibody classes

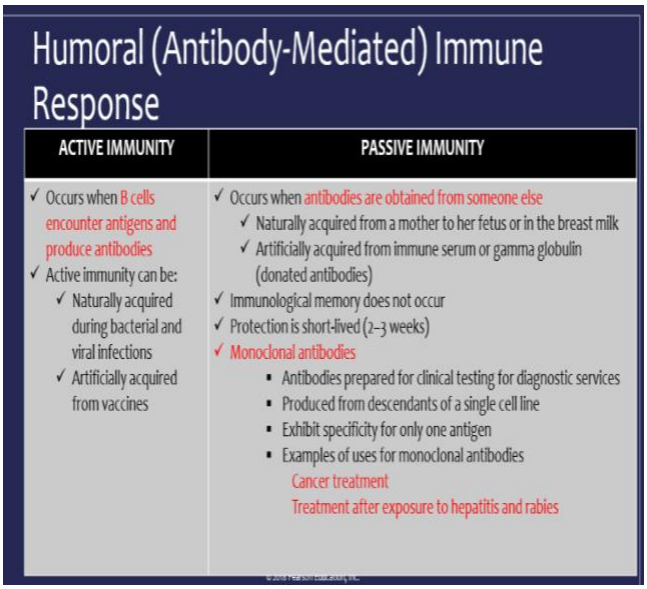
✓ Antibodies of each class have slightly different roles and differ structurally and functionally

✓ Five major immunoglobulin classes (MADGE)

- IgM—can fix complement
- IgA—found mainly in secretions, such as mucus or tears
- IgD—important in activation of B cell
- IgG—can cross the placental barrier and fix complement; most abundant antibody in plasma
- IgE—involved in allergies







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Table 12.2 **Immunoglobulin Classes**

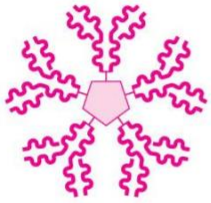
Class	Generalized structure	Where found	Biological function
IgM		Attached to B cell; free in plasma	When bound to B cell membrane, serves as antigen receptor; first Ig class *released* by plasma cells during primary response; potent agglutinating agent; fixes complement.
IgA		Some (monomer) in plasma; dimer in secretions such as saliva, tears, intestinal juice, and milk	Bathes and protects mucosal surfaces from attachment of pathogens.

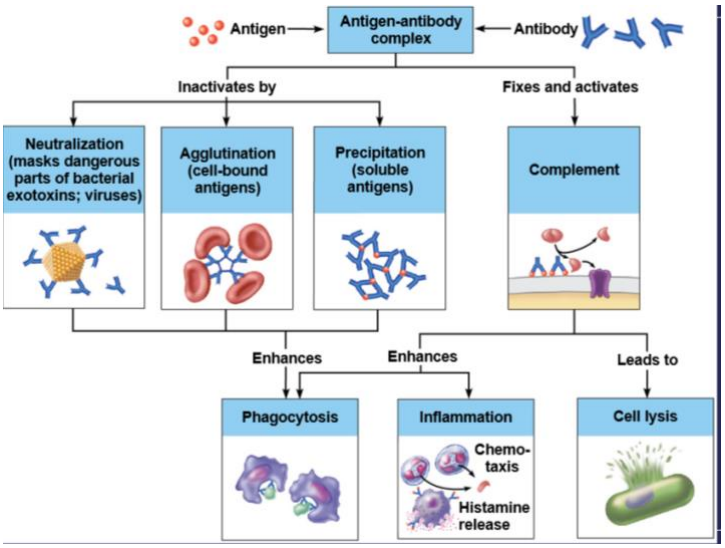
Table 12.2 **Immunoglobulin Classes** (*continued*)

Class	Generalized structure	Where found	Biological function
IgD		Almost always attached to B cell	Believed to be cell surface receptor of immunocompetent B cell; important in activation of B cell.
IgG		Most abundant antibody in plasma; represents 75–85% of circulating antibodies	Main antibody of both primary and secondary responses; crosses placenta and provides passive immunity to fetus; fixes complement.
IgE		Secreted by plasma cells in skin, mucosae of gastrointestinal and respiratory tracts, and tonsils	Binds to mast cells and basophils and triggers release of histamine and other chemical mediators of inflammation and some allergic responses.

❖ Antibody function

✓ Antibodies inactivate antigens in a number of ways

- Complement fixation: chief antibody ammunition used against cellular antigens
- Neutralization: antibodies bind to specific sites on bacterial *exotoxins* or on viruses that can cause cell injury
- Agglutination: antibody-antigen reaction that causes clumping of cells
- Precipitation: cross-linking reaction in which antigen-antibody complex settles out of solution

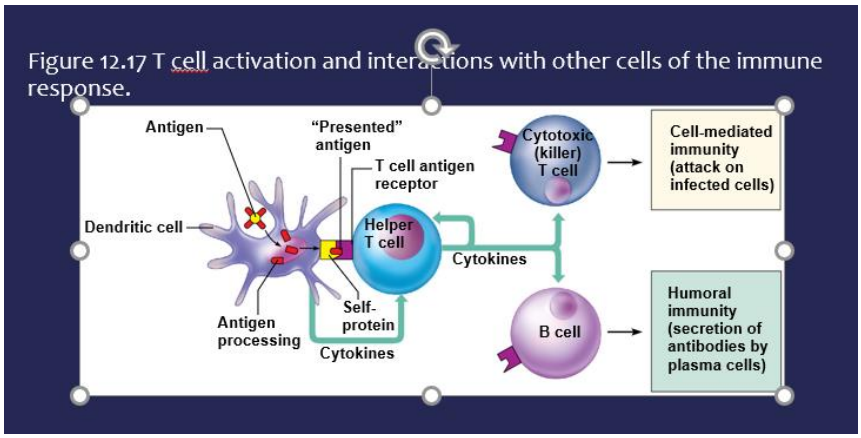


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CELLULAR IMMUNITY = CELL-MEDIATED IMMUNITY

❖ Cytotoxic (killer) T cells

- ✓ Specialize in killing infected cells
- ✓ Insert a toxic chemical (perforin or granzyme)
- ✓ The perforin enters the foreign cell's plasma membrane
- ✓ Pores now appear in the target cell's membrane
- ✓ Granzymes (protein-digesting enzymes) enter and kill the foreign cell
- ✓ Cytotoxic T cell detaches and seeks other targets

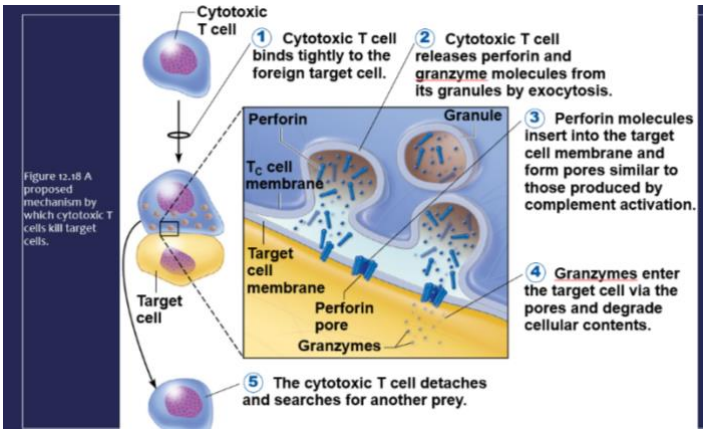


❖ Helper T cells

- ✓ Recruit other cells to fight invaders
- ✓ Interact directly with B cells bound to an antigen, prodding the B cells into clone production
- ✓ Release cytokines, chemicals that act directly to rid the body of antigens

❖ Regulatory T cells

- ✓ Release chemicals to suppress the activity of T and B cells
- ✓ Stop the immune response to prevent uncontrolled activity
- ✓ A few members of each clone are memory cells



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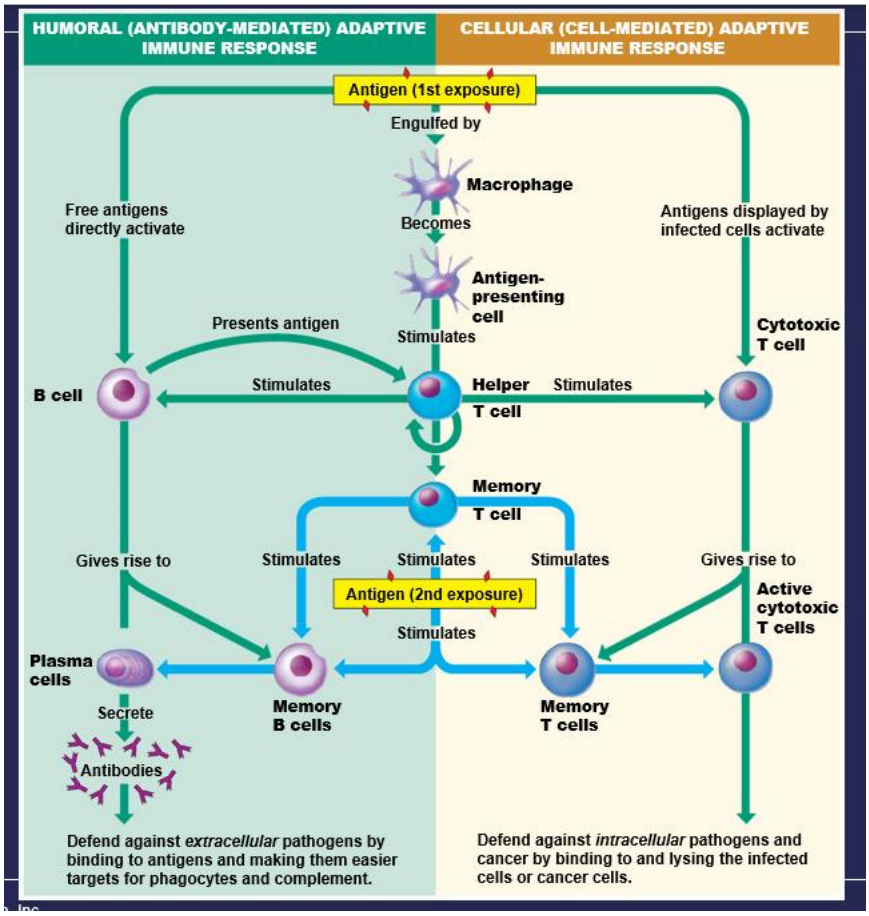


Table **12.3** **Functions of Cells and Molecules Involved in Immunity**

Element	Function in the immune response
Cells	
B cell	Lymphocyte that resides in the lymph nodes, spleen, or other lymphoid tissues, where it is induced to replicate by antigen-binding and helper T cell interactions; its progeny (clone members) form plasma cells and memory cells.
Plasma cell	Antibody-producing "machine"; produces huge numbers of the same antibody (immunoglobulin); specialized B cell clone descendant.
Helper T cell	A T cell that binds with a specific antigen presented by an APC; it stimulates the production of other immune cells (cytotoxic T cells and B cells) to help fight the invader; acts both directly and indirectly by releasing cytokines.

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Table **12.3** **Functions of Cells and Molecules Involved in Immunity** (*continued*)

Element	Function in the immune response
Cytotoxic T cell	Activity enhanced by helper T cells; its specialty is killing virus-invaded body cells, as well as body cells that have become cancerous; involved in graft rejection.
Regulatory T cell	Slows or stops the activity of B and T cells once the infection (or attack by foreign cells) has been conquered. Thought to be important in preventing autoimmune diseases.
Memory cell	Descendant of an activated B cell or T cell; generated during both primary and secondary immune responses; may exist in the body for years thereafter, enabling it to respond quickly and efficiently to subsequent infections or meetings with the same antigen.
Antigen-presenting cell (APC)	Any of several cell types (macrophage, dendritic cell, B cell) that engulfs and digests antigens that it encounters and presents parts of them on its plasma membrane for recognition by T cells bearing receptors for the same antigen; this function, *antigen presentation*, is essential for normal cell-mediated responses. Macrophages and dendritic cells also release chemicals (cytokines) that activate many other immune cells.

Table **12.3** **Functions of Cells and Molecules Involved in Immunity** (*continued*)

Element	Function in the immune response
Molecules	
Antibody (immunoglobulin)	Protein produced by a B cell or its plasma-cell offspring and released into body fluids (blood, lymph, saliva, mucus, etc.), where it attaches to antigens, causing neutralization, opsonization, precipitation, or agglutination, which "marks" the antigens for destruction by phagocytes or complement.
Cytokines	Chemicals released by sensitized T cells, macrophages, and certain other cells: • Migration inhibiting factor (MIF)—"inhibits" macrophage migration and keeps them in the local area. • Interleukin 2—stimulates T cells and B cells to proliferate; activates NK cells. • Helper factors—enhance antibody formation by plasma cells. • Suppressor factors—suppress antibody formation or T cell–mediated immune responses (interleukin-10 transforming growth factor and others). • Chemotactic factors—attract leukocytes (neutrophils, eosinophils, and basophils) into inflamed area. • Gamma interferon—secreted by lymphocytes; helps make tissue cells resistant to viral infection; activates macrophages and NK cells; enhances maturation of cytotoxic T cells.

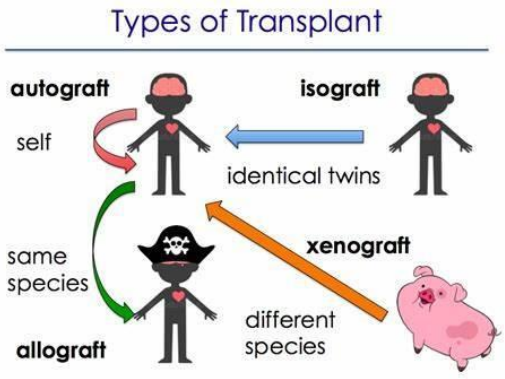
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Table 12.3 **Functions of Cells and Molecules Involved in Immunity** (*continued*)

Element	Function in the immune response
Molecules	
Tumor necrosis factor (TNF)	Like perforin, causes cell killing; attracts granulocytes; activates T cells and macrophages.
Complement	Group of bloodborne proteins activated after binding to antibody-covered antigens; when activated, complement causes lysis of the microorganism and enhances inflammatory response.
Antigen	Substance capable of provoking an immune response; typically a large, complex molecule not normally present in the body.
Cytotoxins	Perforin, granzymes—cell toxins released by cytotoxic T cells and NK cells.

❖ Organ Transplants and Rejection

- Major types of transplants, or grafts
  - ✓ Autografts—tissue transplanted from one site to another on the *same person*
  - ✓ Isografts—tissue grafts from a *genetically identical person* (identical twin)
  - ✓ Allografts—tissue taken from *a person other than an identical twin* (most common type of graft)
  - ✓ Xenografts—tissue taken from a *different animal species* (never successful)
- Blood group and tissue matching is done to ensure the best match possible
- 75% match is needed to attempt a graft
- Organ transplant is followed by immunosuppressive therapy to prevent rejection



❖ Disorders of Immunity

The most important disorders of the immune system

➢ Allergies

- ✓ hypersensitives, are abnormal, vigorous immune responses
- ✓ The immune system overreacts to an otherwise harmless antigen, and tissue damage occurs
- ✓ Types Of Allergies

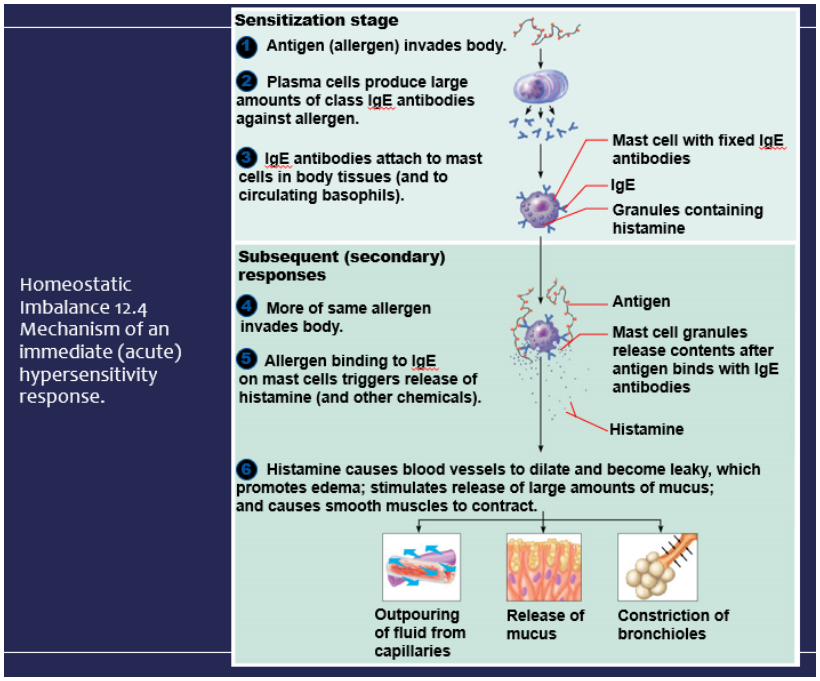
IMMEDIATE (ACUTE) HYPERSENSITIVITY

- Seen in hives and anaphylaxis
- Due to IgE antibodies and histamine
- Anaphylactic shock is systemic, acute allergic response and is rare

DELAYED HYPERSENSITIVITY

- Reflects activity of T cells, macrophages, and cytokines
- Symptoms usually appear 1–3 days after contact with antigen
- Allergic contact dermatitis (poison ivy, cosmetics)

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➢ Autoimmune diseases

✓ Occurs when the body's self-tolerance breaks down

✓ The body produces auto-antibodies and sensitized T lymphocytes that attack its own tissues

✓ Most forms of autoimmune disease result from the appearance of formerly hidden self-antigens or changes in the structure of self-antigens, and antibodies formed against foreign antigens that resemble self-antigens

✓ Examples of autoimmune diseases
- Rheumatoid arthritis—destroys joints
- Myasthenia gravis—impairs communication between nerves and skeletal muscles
- Multiple sclerosis—white matter of brain and spinal cord is destroyed
- Graves' disease—thyroid gland produces excess thyroxine
- Type I diabetes mellitus—destroys pancreatic beta cells, resulting in deficient insulin production
- Systemic lupus erythematosus (SLE)—affects kidney, heart, lung, and skin
- Glomerulonephritis—severe impairment of kidney function due to acute inflammation

➢ Immunodeficiencies

✓ May be congenital or acquired
- Severe combined immunodeficiency disease (SCID) is a congenital disease
- AIDS (acquired immune deficiency syndrome) is caused by a virus that attacks and cripples the helper T cells

✓ Result from abnormalities in any immune element

✓ Production or function of immune cells or complement is abnormal

Developmental Aspects of the Lymphatic System and Body Defenses

✓ Lymphatic vessels form by budding off from veins

✓ Lymph nodes present by fifth week of development

✓ The thymus and the spleen are the first lymphoid organs to appear in the embryo

✓ Other lymphoid organs are poorly developed before birth

✓ The immune response develops around the time of birth

✓ The ability of immunocompetent cells to recognize foreign antigens is genetically determined

✓ Stress appears to interfere with normal immune response

✓ Efficiency of immune response wanes in old age, and infections, cancer, immunodeficiencies, and autoimmune diseases become more prevalent

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